

Effect of sublingual nitroglycerin on absolute coronary blood flow: A randomised double-blind placebo-controlled mechanistic study



Thabo Mahendiran, BMBCh, MD, Adriaan Wilgenhof, MD, Sara Corradetti, MD, Michele Viscusi, MD, Alex Heyse, MD, Tom De Wolf, RN, Nadia Aroudj, RN, Sofie Browaeys, RN, Nancy Claessens, RN, Isabelle Cobbaut, RN, Liesbet De Block, RN, Karolien De Wolf, RN, Hilde Delacroix, RN, Eddy Girukwishaka, RN, Els Mathieu, RN, Katia Van Cutsem, RN, Chloe Van De Velde, RN, Peggy Van Der Kelen, RN, Lyn Verbeken, RN, Ilse De Gend, RN, Isabel De Vulder, RN, Nathalie Haeleydt, RN, Steffi Heremans, MIT, Kathleen Rooselaers, RN, Ayla Verbeurgt, MIT, David Verhelst, RN, Els Van Laer, RN, Olivier Nelis, BMET, Tatyana Storozhenko, MD, Dimitri Buytaert, PhD, Jeroen Sonck, MD, PhD, Nils P. Johnson, MD, MS, Carlos Collet, MD, PhD, Peter Jüni, MD, Bernard De Bruyne, MD, PhD

PII: S1936-8798(26)01320-8

DOI: <https://doi.org/10.1016/j.jcin.2026.04.014>

Reference: JCIN 8772

To appear in: *JACC: Cardiovascular Interventions*

Received Date: 9 March 2026

Revised Date: 14 April 2026

Accepted Date: 21 April 2026

Please cite this article as: Mahendiran T, Wilgenhof A, Corradetti S, Viscusi M, Heyse A, De Wolf T, Aroudj N, Browaeys S, Claessens N, Cobbaut I, De Block L, De Wolf K, Delacroix H, Girukwishaka E, Mathieu E, Van Cutsem K, Van De Velde C, Van Der Kelen P, Verbeken L, De Gend I, De Vulder I, Haeleydt N, Heremans S, Rooselaers K, Verbeurgt A, Verhelst D, Van Laer E, Nelis O, Storozhenko T, Buytaert D, Sonck J, Johnson NP, Collet C, Jüni P, De Bruyne B, Effect of sublingual nitroglycerin on absolute coronary blood flow: A randomised double-blind placebo-controlled mechanistic study, *JACC: Cardiovascular Interventions* (2026), doi: <https://doi.org/10.1016/j.jcin.2026.04.014>.

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no

longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: <https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article>. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2026 Published by Elsevier on behalf of the American College of Cardiology Foundation.

## Randomized Double-Blind Placebo-Controlled Mechanistic Study to Assess Effect of Sublingual NTG on Absolute Coronary Flow

| Study Design                                                                                                                                      | Primary Endpoint                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>40<br/>Patients with angina</p> <p>↓</p> <p>20 Sublingual NTG vs 20 Sublingual placebo</p> <p>↓</p> <p>Measurements at 0, 5 and 10 minutes</p> | <p><b>Absolute coronary blood flow (ml/min)</b><br/>Measured with continuous intracoronary thermodilution in the LAD</p> <p>↓ <b>13.3%</b> (<math>P = 0.001</math>)</p>                                                                                                                                                                       |
|                                                                                                                                                   | <p><b>Secondary Endpoints</b></p> <p><b>Microvascular resistance</b> ↑ <b>8.0%</b> (<math>P = 0.026</math>)</p> <p><b>LVEDP (preload)</b> ↓ <b>38.1%</b> (<math>P &lt; 0.001</math>)</p> <p><b>Aortic pressure (afterload)</b> ↓ <b>8.5%</b> (<math>P &lt; 0.001</math>)</p> <p><b>LV work</b> ↓ <b>21.1%</b> (<math>P &lt; 0.001</math>)</p> |

- Sublingual NTG reduced left ventricular preload and afterload, leading to a marked reduction in myocardial work.
- Despite a significant increase in vessel volume, NTG was associated with an increase in microvascular resistance and a corresponding decrease in absolute coronary blood flow, changes that appear adaptive and secondary to the reduction in myocardial work.

## Effect of sublingual nitroglycerin on absolute coronary blood flow: A randomised double-blind placebo-controlled mechanistic study

Thabo Mahendiran, BMBCh, MD,<sup>1, 2</sup> Adriaan Wilgenhof, MD<sup>1,3</sup>; Sara Corradetti, MD<sup>1</sup>; Michele Viscusi, MD<sup>1</sup> Alex Heyse, MD<sup>1</sup>; Tom De Wolf, RN<sup>1</sup>; Nadia Aroudj, RN<sup>1</sup>; Sofie Browaeys, RN<sup>1</sup>; Nancy Claessens, RN<sup>1</sup>; Isabelle Cobbaut, RN<sup>1</sup>; Liesbet De Block, RN<sup>1</sup>; Karolien De Wolf, RN<sup>1</sup>; Hilde Delacroix, RN<sup>1</sup>; Eddy Girukwishaka, RN<sup>1</sup>; Els Mathieu, RN<sup>1</sup>; Katia Van Cutsem, RN<sup>1</sup>; Chloe Van De Velde, RN<sup>1</sup>; Peggy Van Der Kelen, RN<sup>1</sup>; Lyn Verbeken, RN<sup>1</sup>; Ilse De Gend, RN<sup>1</sup>; Isabel De Vulder, RN<sup>1</sup>; Nathalie Haeleydt, RN<sup>1</sup>; Steffi Heremans, MIT<sup>1</sup>; Kathleen Rooselaers, RN<sup>1</sup>; Ayla Verbeurgt, MIT<sup>1</sup>; David Verhelst, RN<sup>1</sup>; Els Van Laer, RN<sup>1</sup>; Olivier Nelis, BMET<sup>1</sup>; Tatyana Storozhenko, MD<sup>1,4</sup>; Dimitri Buytaert, PhD<sup>1</sup>; Jeroen Sonck, MD, PhD<sup>1</sup>; Nils P. Johnson, MD, MS<sup>5</sup>; Carlos Collet, MD, PhD<sup>1</sup>; Peter Jüni, MD<sup>6‡</sup>; Bernard De Bruyne MD, PhD<sup>1, 2‡</sup>

*‡These authors jointly directed this study*

### Affiliations:

1. Cardiovascular Center Aalst, AZORG, Aalst, Belgium.
2. Department of Cardiology, Lausanne University Hospital, Lausanne, Switzerland.
3. Department of Clinical and Molecular Medicine - Sapienza University, Rome, Italy
4. Department of Prevention and Treatment of Emergency Conditions, L.T. Malaya Therapy National Institute NAMSU, Kharkiv, Ukraine
5. Division of Cardiology, Department of Medicine, McGovern Medical School at University of Texas Health and Memorial Hermann Hospital, Houston, Texas, USA
6. Clinical Trial Service Unit and Epidemiological Studies Unit (CTSU), Nuffield Department of Population Health, University of Oxford, Oxford, United Kingdom

**Running title:** Sublingual nitroglycerin and absolute coronary blood flow

**Total word count:** 4117

## **FUNDING**

None.

## **DISCLOSURE OF INTEREST**

TM is supported by a grant from the Swiss National Science Foundation (SNSF). AW and SC are supported by a research grant from the Digi-CardioPaTh PhD Program, Sapienza University, Rome, Italy. MV reports receiving research grants provided by the Cardiopath Ph.D. program. CC reports receiving research grants from Biosensor, Corovantis Research, Medis Medical Imaging, Pie Medical Imaging, CathWorks, Boston Scientific, Siemens, HeartFlow Inc, Abbott Vascular, and consultancy fees from HeartFlow Inc, OpSens, Abbott Vascular, and Philips Volcano. BDB has a consulting relationship with Boston Scientific, Abbott Vascular, CathWorks, Siemens, and Corovantis Research; receives research grants from Abbott Vascular, Corovantis Research, Cathworks, Boston Scientific; and holds minor equities in Philips-Volcano, Siemens, GE Healthcare, Edwards Life Sciences, HeartFlow, Sanofi, and Celyad. The remaining authors have nothing to disclose.

## **Address for correspondence:**

Bernard De Bruyne, MD, PhD

Cardiovascular Center Aalst, AZORG

Moorselbaan 164,

9300 Aalst, Belgium

## **ABSTRACT**

### **Background**

Nitrates remain a key drug class in the management of angina pectoris. However, their effect on coronary blood flow has never been convincingly demonstrated in humans.

### **Objectives**

We assessed the impact of sublingual nitroglycerin (NTG) on absolute coronary blood flow, coronary resistance, and left ventricular work.

### **Methods**

We conducted a randomised, double-blind, placebo-controlled, mechanistic study. Forty patients with suspected angina underwent cardiac catheterisation and randomisation to sublingual NTG (n=20) or a placebo control (n=20). Using continuous intracoronary thermodilution, baseline absolute coronary blood flow (primary outcome) and a range of other haemodynamic parameters were measured. Following delivery of the sublingual intervention, repeat measurements were made after 5 and 10 minutes. Between-group differences were analysed using analysis of covariance.

### **Results**

Sublingual NTG resulted in a significant decrease in mean LVEDP by 38.1% (95% CI 21.8 to 54.4) (estimated mean difference: -5.4 mmHg, 95% CI -7.8 to -3.1,  $p<0.001$ ), along with a significant decrease in aortic pressure by 8.5% (95% CI 5.3 to 11.7) (estimated mean difference: -8.6 mmHg, 95% CI -12.2 to -5.1,  $p<0.001$ ) and distal coronary pressure by 7.4% (95% CI 3.7 to 11.1) (estimated mean difference: -7.1 mmHg, 95% CI -10.6 to -3.7,  $p<0.001$ ) after 10 minutes. Myocardial work also decreased by 21.1% (95% CI 15.4 to 26.9) (estimated

mean difference: -8.9 mmHg•s, 95% CI -11.6 to -6.2,  $p<0.001$ ). Despite inducing a significant increase in vessel volume, NTG resulted in a significant increase in microvascular resistance by 8.0% (95% CI 1.1 to 14.9) (estimated mean difference: 121.5 Wood units, 95% CI 15.3 to 227.8,  $p=0.026$ ), and a corresponding decrease in absolute coronary blood flow by 13.3% (95% CI 6.4 to 20.2) (estimated mean difference: -8.7 ml/min, 95% CI -13.4 to -3.9,  $p=0.001$ ).

### **Conclusion**

Sublingual NTG induced a significant decrease in left ventricular preload and afterload, resulting in a marked decrease in myocardial work. This was associated with a significant and adaptive decrease in absolute coronary blood flow.

**Keywords:** nitroglycerin (NTG), angina, coronary artery disease, absolute coronary flow, continuous intracoronary thermodilution

**CONDENSED ABSTRACT**

Nitrates remain central to angina management, yet their effect on coronary blood flow in humans is uncertain. In a randomised, double-blind, placebo-controlled mechanistic study, 40 patients with suspected angina underwent continuous intracoronary thermodilution before and after sublingual nitroglycerin (NTG) or placebo. NTG significantly reduced LVEDP, aortic and distal coronary pressures, and myocardial work. Despite increasing vessel volume, NTG increased microvascular resistance and decreased absolute coronary blood flow by 13.3%. These findings indicate that sublingual NTG reduces preload and afterload, lowering myocardial work and producing an adaptive decrease in absolute coronary blood flow.



**ABBREVIATIONS**

NTG = nitroglycerin

LVEDP = left ventricular end-diastolic pressure

$P_a$  = aortic pressure

$P_d$  = distal coronary pressure

$Q$  = absolute coronary flow

$R_\mu$  = microvascular resistance

SPTI = systolic pressure-time index

$T$  = temperature of mixed blood and saline in the distal vessel

$T_i$  = entry temperature of saline

## INTRODUCTION

Nitrates remain a key drug class in the management of angina pectoris.<sup>1</sup> Their principal mechanism of action relates to the relaxation of vascular smooth muscle, leading to vasodilation.<sup>2</sup> This vasodilatory capacity has multiple targets that are believed to lead to their antianginal effect.

First, nitrates are predominantly venodilators, resulting in decreased cardiac preload, left ventricular diastolic pressure, and myocardial wall stress.<sup>3</sup> Second, nitrates trigger arterial and arteriolar dilatation at low doses, eventually reducing systemic vascular resistance and cardiac afterload at higher doses.<sup>4</sup> Third, nitrates also directly affect the coronary vasculature, inducing vasodilatation of the epicardial arteries and arterioles with a diameter  $>100\text{ }\mu\text{m}$ .<sup>1</sup> The coronary microcirculation, despite being resistant to this vasodilatory effect, is also impacted – albeit indirectly – as the decreased myocardial wall stress induced by systemic venous dilatation has been shown to increase flow from the epicardium to the endocardium.<sup>5</sup>

Yet, despite our understanding of the effects of nitrates on cardiac filling pressures, systemic vascular resistance, and the coronary circulation, to the best of our knowledge, their impact on absolute coronary blood flow has never been studied in humans in a randomised double-blind placebo-controlled study, in part due to the absence of reliable techniques to quantify it. The complex interaction between many of the factors influenced by nitrates makes their overall effect on coronary flow challenging to appreciate, even from a theoretical standpoint. This further highlights an important gap in our understanding of this fundamental cardiovascular drug.

Continuous intracoronary thermodilution now permits the precise and accurate measurement of absolute coronary flow in humans.<sup>6</sup> This extensively validated technique<sup>7-12</sup> has opened the door to previously inaccessible physiological studies.<sup>13,14</sup>

To this end, we performed a randomised double-blind, placebo-controlled mechanistic study using continuous intracoronary thermodilution to assess the impact of sublingual nitroglycerin (NTG) on absolute coronary blood flow, absolute epicardial and microvascular resistance, and haemodynamic parameters that determine myocardial work.

## METHODS

### Patient population

We included patients aged  $\geq 18$  years referred for invasive coronary angiography (ICA) for suspected coronary artery disease. Patients were excluded if: they had  $\geq 1$  severe stenosis (defined as a diameter stenosis  $> 70\%$ ), clinical signs or symptoms of congestive heart failure, severe valvular heart disease requiring either surgical or percutaneous intervention, a history of coronary artery bypass grafting, tortuous coronary anatomy in which wire manipulation could be complex, significant arrhythmia, use of vasoactive medications within 24 hours of ICA. The study protocol was approved and registered with the IRB of the OLV Clinic Aalst, Belgium (study number: OLV2024/037; B1262024000004; ClinicalTrials.gov ID: NCT06823843), and all patients provided informed consent.

### Outcomes

The primary outcome was a change in absolute coronary flow from baseline, 10 minutes after administration of the intervention, with an intermediate measurement also made after 5 minutes. The secondary outcomes were changes in the following variables over the same timeframe: microvascular resistance, epicardial resistance, aortic pressure ( $P_a$ ), distal coronary pressure ( $P_d$ ), systolic pressure-time index (SPTI), and the Buckberg index.

### Study protocol

#### *(i) Coronary angiography*

Coronary angiography was performed via the radial or femoral artery using a 6 Fr guiding catheter. Critically, in contrast to routine clinical practice, no vasoactive medications were used before or during diagnostic angiography, including no intra-arterial vasodilators for the prevention of radial spasm and no intracoronary isosorbide dinitrate. The LVEDP was

measured using a pigtail catheter before injection of the coronary arteries. Upon completion of diagnostic angiography, each patient underwent randomisation. Pullback of the pigtail catheter from the left ventricular (LV) to the ascending aorta permitted the exclusion of a significant aortic valve gradient.

*(ii) Randomisation and blinding*

Patients were randomised in a 1:1 ratio to either sublingual NTG (2 puffs for a total dose of 800 µg) or placebo (2 puffs of sublingual saline). The selected NTG dose is typically administered for the treatment of angina attacks. In addition, it is the recommended dose before coronary CT angiography to improve vessel visualisation and stenosis assessment as it induces maximal epicardial vasodilatation.<sup>15,16</sup>

The allocation sequence was computer-generated using a single block of 40. Randomisation was performed by a dedicated study nurse who was not involved in patient recruitment using sequentially numbered, sealed, opaque envelopes, which were stored securely and accessed only at the time of allocation.<sup>17</sup> The NTG and placebo interventions were in identical bottles indistinguishable in shape, size, and colour. After patient inclusion, the study nurse opened the envelope with the next sequential number and administered the assigned intervention in accordance with the allocation schedule. Patients, operators, and all other clinical team members were blinded to group allocation.

*(iv) Continuous intracoronary thermodilution*

A guidewire equipped with a pressure/temperature sensor (PressureWire X; Abbott) was connected to dedicated software for tracing visualisation and analysis (CoroFlow Cardiovascular System; Coroventis) and, after zeroing, was advanced through the guiding

catheter. The pressures recorded by the pressure/temperature wire and by the fluid-filled guide catheter were equalised close to the tip of the guiding catheter. The wire was advanced into the distal part of the left anterior descending artery (LAD), and the temperature was zeroed.

For the measurement of absolute coronary flow measurement, a dedicated monorail infusion 2.52 Fr microcatheter with four distal side holes (RayFlow; Hexacath, Paris, France) was advanced over the pressure/temperature wire and connected to the 200 ml motorised syringe of an automated injection system (Medrad Stellant; Medrad Inc. [now Bayer]) filled with room temperature saline. The infusion catheter was advanced into the LAD with its tip positioned within the first millimetres of the artery. Further details of continuous intracoronary thermodilution have been described elsewhere.<sup>12</sup>

*(iv) Measurement of coronary flow, distal coronary pressure, and aortic pressure*

Critically, no vasoactive medications were given before or during diagnostic angiography, and any pre-existing vasoactive medications were stopped 24 hours before the protocol. Upon completion of diagnostic angiography, baseline absolute coronary blood flow ( $Q_{rest}$ ) was measured using continuous intracoronary thermodilution for 3 minutes, before stopping the saline infusion and administering the sublingual intervention (**Figure 1**). Resting flow measurements were then repeated at both 5 minutes and 10 minutes after the sublingual intervention. This measurement timing was informed by pharmacokinetic data indicating that sublingual NTG reaches peak plasma concentrations at approximately 5 minutes after administration, but as late as 10 minutes in some individuals.<sup>18</sup> Importantly, the pressure and temperature tracings were recorded uninterrupted throughout the whole protocol (**Figure 2**).

All resting flow measurements were performed using a saline infusion rate of 10 ml/min, as per current recommendations.<sup>12</sup>

Following the last flow measurement, the pressure wire and infusion catheter were removed, and two injections of the left coronary artery were made using the same projection angles as during baseline angiography. Finally, the LVEDP was remeasured at 12 minutes following removal of the guiding catheter and reinsertion of the pigtail catheter in the LV (**Figure 1**). Two example tracings are shown in **Figure 2**. Examples of left ventricular pressure tracings are shown in **Supplementary Figure 1**.

Pressure and temperature drift were rigorously controlled throughout the protocol. Verification of zero pressure against atmospheric pressure was performed at the end of the protocol, with no significant pressure drift observed in any patient. All temperature measurements were adjusted to account for drift during the protocol. Overall, temperature drift was minimal, with a mean drift of  $0.015 \pm 0.03$  °C in the study population, and no significant differences between groups ( $p = 0.45$ ).

Crucially, absolute coronary blood flow measured by continuous thermodilution is not influenced by changes in epicardial vessel size, making it an ideal technique for the present study. Unlike Doppler or bolus thermodilution techniques, which estimate flow indirectly from blood velocity and are thus heavily affected by vessel cross-sectional area (CSA) as per the continuity equation (flow = CSA x velocity), continuous thermodilution directly quantifies volumetric flow (ml/min) based on the dilution of infused saline within the blood.

*(v) Pressure-wire-based assessment of epicardial disease*

Epicardial disease was assessed using the baseline  $P_d/P_a$  ratio and the diastolic pressure ratio (dPR), defined as the mean  $P_d$  to mean  $P_a$  ratio during the entire diastolic phase of the cardiac cycle.<sup>19</sup> It was extracted programmatically from the baseline resting pressure tracings with the mean calculated from five consecutive cardiac cycles (cut-off value is  $\leq 0.89$ ).<sup>20</sup>

### Data extraction and analysis

Raw waveform data (100 Hz signals) were exported from Coroflow (Coroventis) for analysis. For each 2-minute measurement period, the final minute was averaged to obtain a mean  $P_a$ ,  $P_d$ , heart rate, and mixed blood and saline temperature (T). This corresponded to the minute before the sublingual intervention (baseline  $Q_{rest}$ ), the fifth minute (repeat  $Q_{rest}$  1), and the tenth minute (repeat  $Q_{rest}$  2) (**Figure 1**). The extracted  $P_a$  data were also used to compute the integral of the systolic period programmatically and, thus, the systolic pressure-time index (see below).

### Coronary flow and resistance indices

Absolute resting coronary flow ( $Q_{rest}$ ) in ml/min was calculated as per **Equation 1**, where  $T_i$  is the temperature of the infusate at the tip of the catheter, T is the temperature of blood in the distal vessel, and  $Q_i$  is the saline infusion rate. Importantly, the values of T and  $T_i$  are relative to blood temperature. To simplify the protocol,  $T_i$  was predicted using a validated equation.<sup>21</sup>

$$Q_{rest} = 1.08 \cdot \frac{T_i}{T} \cdot Q_i$$

Where  $T_i$  was predicted using:

$$T_i = 0.98T - 0.28Q_i + 0.90$$



Absolute resting microvascular resistance ( $R_{\mu,rest}$ ) in Wood Units (WU) was calculated using **Equation 2**, where  $P_{d,rest}$  is distal coronary pressure at rest:

$$R_{\mu,rest} = \frac{P_{d,rest}}{Q_{rest}}$$

Absolute resting epicardial resistance ( $R_{epi,rest}$ ) in Wood Units (WU) was calculated using **Equation 3**, where  $P_{a,rest}$  is the aortic pressure at rest:

$$R_{epi,rest} = \frac{P_{a,rest} - P_{d,rest}}{Q_{rest}}$$

### **Preload and afterload**

Left ventricular preload represents the degree of stretching of the cardiac myocytes before contraction, with LVEDP representing an established surrogate measure.<sup>22</sup> Left ventricular afterload – the pressure against which the left ventricle must work during systole – was estimated using  $P_a$ .<sup>23</sup>

### **Myocardial work and perfusion**

#### *(i) Systolic pressure-time index (SPTI)*

The SPTI was extracted programmatically from the raw  $P_a$  pressure curves. It was calculated as the area under the pressure-time curve from the onset of systole to the dicrotic notch (i.e. the integral), with the mean value calculated for five consecutive beats at the end of each measurement period. The SPTI reflects myocardial oxygen demand.<sup>24</sup> Whilst the SPTI is normally calculated from the LV pressure curve, in the absence of aortic outflow obstruction, it can also be estimated from the area under the  $P_a$  curve.<sup>25</sup> In addition, given that myocardial

work is also governed by heart rate, the SPTI was multiplied by heart rate to estimate LV work per minute (LVW/min). This provided a measure of myocardial work that accounted for any changes in heart rate during the protocol.

### *(ii) Buckberg index*

The Buckberg index is a valuable tool for the assessment of myocardial perfusion, representing the ratio between myocardial oxygen supply and demand. In the present study, it was used to assess changes in myocardial oxygen supply and demand following the administration of sublingual NTG. As previously reported,<sup>26</sup> this was calculated by dividing the diastolic pressure-time index (i.e. time of coronary perfusion) by the systolic pressure-time index (i.e. time of myocardial oxygen demand).

**Supplementary Figure 2** shows example patient tracings with these indices computed.

### **3D quantitative coronary angiography**

Offline 3D-QCA using QAngio XA 3D software (Medis, Leiden, The Netherlands) of the LAD was performed both at baseline and after the completion of flow measurements. Two projections at 15 frames per second, differing by more than 30° angulation, were required for the 3D-QCA. The entire LAD was visible in both projections, and foreshortening of the artery was avoided as much as possible. The ostium of the LAD was chosen as the proximal landmark, and the distal landmark was an easily identifiable distal anatomical marker (e.g. branch point) before the vessel decreased below a diameter of <2 mm. A blinded assessor performed all measurements. Vessel volumes were computed at baseline and post-intervention. For comparison purposes, vessel areas/volumes were also computed using an open-source solution (*AngioPy*).<sup>27</sup>

### Statistical analyses

A sample size of 40 patients, randomly assigned in a 1:1 ratio (20 per group), provided more than 80% power to detect a between-group difference in change from baseline in absolute resting coronary flow in the LAD at 10 minutes of 21.0 ml/min at a two-sided alpha of 0.05, assuming no change in the placebo group and a 30% increase from  $69.8 \pm 23.0$  ml/min at baseline to  $90.8 \pm 23.0$  ml/min at 10 minutes in the NTG group. These assumptions were based on a recently reported mean and standard deviation for absolute coronary flow in the LAD of  $69.8 \pm 23.0$  ml/min.<sup>13</sup> As no data were available on the impact of NTG on absolute coronary flow in humans, we relied on a canine model in which administration of 1.2 mg of sublingual NTG led to an approximate 50% increase in late diastolic coronary blood flow at 3 minutes.<sup>28</sup> Taking into account the lower sublingual NTG dose used in humans, we assumed a more conservative 30% increase from baseline in the NTG group, corresponding to an increase in absolute blood flow of 21.0 ml/min. A post hoc sample size consideration, based on the observed pooled standard deviation of absolute coronary flow of 16.1 ml/min at 10 minutes and the observed correlation between baseline and 10 minutes of 0.855, indicated that an analysis of covariance (ANCOVA) would provide 80% power to detect a between-group difference in absolute coronary flow of 7.4 ml/min in either direction at a two-sided alpha of 0.05. Under these parameters, the corresponding minimal detectable difference, defined as the minimal between-group difference in absolute coronary flow that would yield statistical significance, was 5.2 ml/min.

All outcomes were analysed using analysis of covariance adjusted for the outcome's baseline values, and are reported as differences in means with 95% confidence intervals and corresponding two-sided P values. To facilitate interpretability, between-group differences

were also expressed as a percentage change and compared using a t-test. All randomised participants were included in the analyses according to their randomised allocation, and there were no missing outcome data. No multiplicity adjustment was used; the widths of the 95% confidence intervals and the p-values for secondary outcomes were not adjusted for multiplicity and should be considered exploratory. Continuous variables were found to be approximately normally distributed and are summarised as mean  $\pm$  standard deviation; categorical variables are summarised as counts and percentages. P-values  $<0.05$  were considered statistically significant. All analyses were performed using Python 3.11.4 (Python Software Foundation, Wilmington, DE, USA) and R software, version 4.11 (R Foundation for Statistical Computing, Vienna, Austria).

## RESULTS

### (i) Study population

A total of 40 patients were randomised, 20 to placebo and 20 to NTG, all of whom were included in the final analysis. The study was completed safely without any complications. None of the patients receiving NTG reported any symptoms related to its administration. Baseline characteristics were similar between groups (**Table 1**). Overall, the mean age of the population was  $65.7 \pm 10.3$  years, with the majority being male (77.5%). Chest pain was the most frequent indication for coronary angiography (65%). Of note, there was no difference in epicardial disease between groups, both with regard to anatomical severity (mean diameter stenosis: placebo:  $41.7 \pm 10.4\%$  vs NTG:  $39.0 \pm 8.8\%$ ) and haemodynamic severity (mean  $P_d/P_a$ : placebo  $0.90 \pm 0.03$  vs NTG:  $0.89 \pm 0.09$ ; mean dPR: placebo:  $0.84 \pm 0.07$  vs NTG:  $0.85 \pm 0.11$ ). None of the patients had significant aortic valve disease, as confirmed invasively by a mean peak-to-peak gradient of  $3.1 \pm 2.8$  mmHg.

### (ii) Effect of sublingual NTG on left ventricular preload

Compared with placebo, the administration of NTG was associated with a significant decrease in mean LVEDP, with an estimated mean difference of -5.4 mmHg after 12 minutes (95% CI -7.8 to -3.1,  $p < 0.001$ ), which corresponded to a percentage difference of -38.1% (95% CI -54.4 to -21.8; **Figure 3**).

### (iii) Effect of sublingual NTG on left ventricular afterload

NTG was associated with a significant sequential decrease in  $P_a$  compared with placebo with an estimated mean difference of -4.7 mmHg (95% CI -7.5 to -2.0,  $p = 0.001$ ) after 5 minutes and -8.6 mmHg (95% CI -12.2 to -5.1,  $p < 0.001$ ) at 10 minutes (10-minute percentage difference: -8.5%, 95% CI -11.7 to -5.3; **Figure 4A**). In addition, there was a corresponding

sequential decrease in  $P_d$  with an estimated mean difference of -3.6 mmHg (95% CI -6.0 to -1.1,  $p=0.006$ ) after 5 minutes and -7.1 mmHg (95% CI -10.6 to -3.7,  $p<0.001$ ) after 10 minutes (10-minute percentage difference: -7.4%, 95% -11.1 to -3.7; **Figure 4B**).

#### **(iv) Effect of sublingual NTG on myocardial work and perfusion**

##### *Systolic pressure-time index (SPTI)*

NTG resulted in a significant sequential decrease in myocardial work as compared with placebo with an estimated mean difference of -5.8 mmHg•s (95% CI -8.0 to -3.7,  $p<0.001$ ) after 5 minutes and -8.9 mmHg•s (95% CI -11.6 to -6.2,  $p<0.001$ ) after 10 minutes, corresponding to a 10-minute percentage difference of -21.1% (95% CI -26.9 to -15.4) (**Figure 5A**).

Estimated LV work per minute (mmHg•s/minute) also decreased significantly confirming that these findings were consistent after accounting for changes in heart rate during the study (estimated mean difference at 5 minutes: -319.8 mmHg•s/min (95% CI -516.7 to -122.9,  $p=0.002$ ); estimated mean difference at 10 minutes: -515.8 mmHg•s/min (95% CI -725.3 to -306.3,  $p<0.001$ ; 10-minute percentage difference: -18.5%, 95% CI -25.2 to -11.8; **Figure 5B + Supplementary Figure 3**).

##### *Buckberg index*

NTG was associated with an improvement in myocardial perfusion as defined by the Buckberg index, with an estimated mean difference of 0.2 (95% CI 0.1 to 0.3,  $p=0.001$ ) after 5 minutes. This difference persisted at 10 minutes (estimated mean difference 0.2, 95% CI 0.1 to 0.3,  $p<0.001$ ; 10-minute percentage difference: 17.7%, 95% CI 9.3 to 26.1; **Figure 5C**).

**(iv) Effect of nitrates on coronary artery size, epicardial resistance, and microvascular resistance**

NTG induced a significant increase in vessel volume by 15.6% (95% CI 0.9 to 30.2) at 10 minutes compared with placebo (estimated mean difference 64.1 mm<sup>3</sup>, 95% CI 1.1 to 127.2,  $p=0.046$ ). However, there was no difference in epicardial resistance between groups (estimated mean difference at 5 minutes: -5.7 WU, 95% CI -17.0 to 5.7,  $p=0.32$ ; 10 minutes: -3.2 WU, 95% CI -18.6 to 12.1,  $p=0.67$ ; **Supplementary Figure 4**).

NTG resulted in a significant increase in  $R_{\mu,rest}$  following NTG administration compared with placebo, with an estimated mean difference of 65.7 WU (95% CI -47.5 to 178.8,  $p=0.25$ ) after 5 minutes and 121.5 WU (95% CI 15.3 to 227.8,  $p=0.026$ ) after 10 minutes (10-minute percentage difference: 8.0%, 95% CI 1.1 to 14.9; **Figure 6A**).

**(v) Effect of nitrates on absolute resting coronary flow (primary outcome)**

NTG resulted in a significant decrease in absolute coronary flow compared with placebo, with an estimated mean difference of -4.7 ml/min (95% CI -9.6 to 0.1,  $p=0.054$ ) after 5 minutes and -8.7 ml/min (95% CI -13.4 to -3.9,  $p=0.001$ ) after 10 minutes, corresponding to a 10-minute percentage difference of -13.3 % (95% CI -20.2 to -6.4; **Figure 6B**). This was associated with a significant increase in the  $P_d/P_a$  ratio in the NTG group compared with placebo, with a mean difference of 0.01 (95% CI 0.00 to 0.02,  $p=0.026$ ) and a 10-minute percentage difference of 1.2% (95% CI -0.1 to 2.6).

A summary of all results is shown in **Table 2**. Patient-level plots for all reported parameters are provided in the **Supplementary Appendix**.

A summary of the major study findings is shown in the **Central Illustration**.

Journal Pre-proof



## DISCUSSION

To the best of our knowledge, this is the first randomised double-blind placebo-controlled mechanistic study that assesses the impact of sublingual nitroglycerin on absolute coronary blood flow, epicardial and microvascular resistance, and haemodynamic parameters that determine myocardial work. Contrary to our initial assumptions, which were informed by a preclinical study published in 1978,<sup>28</sup> sublingual NTG, compared to placebo, was associated with a 13.3% decrease in absolute resting coronary flow (primary outcome). This reduction was secondary to significant decreases in both left ventricular preload and afterload, resulting in a marked decrease in myocardial work. These changes were evident after 5 minutes but became more pronounced after 10 minutes. Whilst sublingual NTG induced significant epicardial vasodilatation, there was no significant change in epicardial resistance.

These results provide the first clear evidence in humans of the decrease in absolute coronary blood flow induced by sublingual NTG, challenging a commonly held perception that NTG relieves angina, in part, by increasing absolute coronary flow. Instead, our findings support a mechanism predominantly driven by a reduction in myocardial oxygen demand, with a corresponding, likely adaptive, decrease in coronary flow. Importantly, beyond these specific observations, our study provides a more comprehensive mechanistic understanding of the coronary and haemodynamic effects of nitrates. We believe that such mechanistic insight remains of fundamental importance in clinical practice, particularly for a drug that is so widely used, as it enables a more informed interpretation of both changes in symptoms and cardiovascular physiology following its administration.

### **Effects on left ventricular preload, afterload, and work**

Nitrates induce vasodilatation by stimulating soluble guanylate cyclase. The resulting increase in cyclic guanosine monophosphate (cGMP) triggers the extrusion of calcium from smooth muscle cells and vascular relaxation.<sup>1</sup> Whilst nitrates dilate veins, arteries, and arterioles, veins appear to take up nitrates more avidly than other vessels. This pronounced venodilation increases venous capacitance, thus triggering the redistribution of blood away from the heart and lungs to the splanchnic and peripheral circulation.<sup>2</sup> The resultant decrease in venous return explains the decrease in left ventricular preload (as measured by LVEDP) associated with nitrates.<sup>3</sup> In the present study, we demonstrated a marked decrease in LVEDP of 38.1% following the administration of sublingual NTG.

Sublingual NTG also induces peripheral arterial vasodilation leading to a well-documented reduction in systemic vascular resistance and blood pressure.<sup>3,29-31</sup> The reduced cardiac preload and thus stroke volume (as per the Frank-Starling law) induced by NTG may also contribute to this decrease in systemic blood pressure. Our data demonstrated a NTG-induced decrease in  $P_a$  by 8.5% after 10 minutes.

The decreases in left ventricular preload and afterload result in a reduced left ventricular end-diastolic volume.<sup>2</sup> This resultant reduction in both systolic and diastolic left ventricular wall stress has been shown to lower myocardial energy demands.<sup>31-33</sup> We observed a progressive and marked reduction in myocardial work by 21.1% after the administration of NTG.

Furthermore, we noted favourable changes to the myocardial oxygen supply and demand ratio, with an increase in the Buckberg index by 17.7%. This increase in the ratio of diastolic-to-systolic time theoretically increases the proportion of time in the cardiac cycle during which the coronary arteries are perfused.<sup>25</sup>

### Effects on the coronary circulation

Our results confirm the vasodilatory effect of nitrates on the epicardial vessels, in line with recently reported data.<sup>16</sup> However, this was not accompanied by a significant change in epicardial resistance in this cohort of patients with mild-to-moderate epicardial disease. This observation should be interpreted with caution, as the study was likely underpowered to detect small but potentially clinically relevant differences in epicardial resistance. Notably, mean epicardial resistance accounted for little more than 10% of total coronary resistance in this study, with the majority attributable to the microcirculation, underscoring its dominant role in determining absolute coronary flow.<sup>34</sup>

Previous studies on the impact of NTG on coronary flow have yielded conflicting results. As far back as 1910, Bond et al. suggested that coronary flow decreased with NTG in a canine model, following a course that was parallel to the changes seen in blood pressure.<sup>35</sup> However, in a later canine experiment by Vatner et al., sublingual NTG appeared to be associated with an increase in coronary blood velocity as measured by Doppler.<sup>28</sup> Limited data exist assessing absolute coronary flow in humans, with previous studies tending to assess myocardial blood flow i.e. perfusion per gram of myocardium, as opposed to absolute coronary blood flow. Bernstein et al. utilised a radioisotope technique to study myocardial blood flow and, whilst identifying decreases in aortic pressure and cardiac output, failed to find any significant decrease in myocardial blood flow both in a canine model and in human subjects.<sup>33</sup> A similar experimental approach by Gorlin et al. also failed to show convincing results on coronary flow.<sup>31</sup> Bernstein et al. hypothesised that the systemic effects of the drug potentially masked its direct effects on the coronary circulation.<sup>33</sup> However, more likely, this reflected limitations in study methodology and sample size.

In the present study, the use of continuous intracoronary thermodilution permitted the precise and accurate measurement of absolute coronary blood flow. Crucially, our experimental setup permitted the objective assessment of the multiple effects of sublingual NTG on left ventricular haemodynamics and work, and their combined effect on absolute coronary flow. We observed an approximate 13.3% reduction in absolute coronary flow 10 minutes after NTG administration. This finding was consistent with the other important haemodynamic effects induced by NTG. We hypothesise that the marked decrease in both left ventricular preload and afterload, and the consequential decrease in left ventricular work, led to an appropriate decrease in coronary flow to match the reduced metabolic requirements. The reduction in flow was also reflected in the small but significant increase in the  $P_d/P_a$  ratio in the NTG group (as explained by the Hagen-Poiseuille equation).<sup>34</sup> This reduction in absolute coronary blood flow induced by NTG should be interpreted as an adaptive and entirely physiological response. Rather than representing true “autoregulation” - defined as the maintenance of stable blood flow despite fluctuations in myocardial perfusion pressure, provided metabolic demands remain constant<sup>14</sup> - it more accurately reflects appropriate “regulation” of coronary flow in response to changing metabolic demands. With respect to the underlying mechanism, although changes in perfusion pressure do influence coronary flow, the principal determinant remains the dynamic modulation of microvascular resistance within the coronary microcirculation.<sup>34</sup> This concept is further illustrated by our recent work, which demonstrates that patients presumed to be at rest can nonetheless exhibit marked fluctuations in absolute coronary flow, driven by rapid, beat-to-beat adjustments in microvascular tone to match myocardial metabolic requirements.<sup>36</sup> Interestingly, in the present study, the favourable increase in the Buckberg index seen with NTG suggests that decreased coronary flow was also partially offset by a relatively longer diastolic perfusion phase.

**Effect of sublingual NTG during rest and exercise**

We hypothesise that the haemodynamic effects of NTG – particularly reductions in preload and afterload – also occur during exercise, as these mechanisms are not specific to rest. Supporting this, Parker et al. found that topical NTG was associated with a significant decrease in LVEDP in patients both with and without coronary artery disease.<sup>37</sup> Thus, a reduction in preload during exercise by sublingual NTG would also be expected.

In addition, Asrress et al. demonstrated a significant decrease in aortic pressure at peak exercise following NTG administration, along with a reduction in SPTI, suggesting that the effects on afterload and left ventricular work observed in our study likely extend to hyperaemia. Yet, the same authors found no significant change in flow velocity or microvascular resistance as assessed by intracoronary Doppler. However, we believe that these findings should be interpreted with caution, as Doppler assesses flow velocity rather than volumetric flow ( $Q = \text{cross-sectional area} \times \text{velocity}$ ). Given that both nitrates and hyperaemia induce epicardial vasodilatation, velocity cannot be interpreted in isolation. Our study addresses this limitation by using continuous intracoronary thermodilution to directly quantify absolute volumetric flow. Unlike velocity-based techniques, this method is not influenced by changes in epicardial vessel size, allowing accurate assessment of coronary blood flow across all phases of the protocol, independent of vessel tone.

With respect to flow during hyperaemia, we speculate that, given the reduction in major determinants of left ventricular work and myocardial oxygen demand, a corresponding decrease in absolute coronary flow may also occur. However, this hypothesis requires

evaluation in a separate randomised study using continuous thermodilution under exercise conditions.

### **The anti-anginal effect of sublingual NTG**

Angina is a manifestation of myocardial ischaemia arising from an imbalance between oxygen supply and demand. Our findings suggest that the anti-anginal mechanism of sublingual NTG results from a significant decrease in myocardial work and thus oxygen demand. This is coupled with more optimised coronary perfusion that permits an appropriate decrease in absolute coronary flow. Placed into the clinical context of angina, NTG likely decreases myocardial work and oxygen demand sufficiently to move patients with a flow-limiting stenosis away from the ischaemic threshold. Of note, NTG has also been shown to modulate the distribution of blood flow and thus improve the perfusion of ischemic zones.<sup>38,39</sup> However, the exploration of changes in regional and transmural blood flow was beyond the scope of this work.

### **Limitations**

First, this study only aimed to study the cardiovascular effects of sublingual NTG under resting conditions. The addition of a hyperaemic or exercise phase to the protocol was beyond the scope of this already complex protocol. However, we believe that the physiological changes induced by NTG in the resting state are also applicable to the hyperaemic state. Second, given that the effects of NTG appeared to progress between 5 and 10 minutes, the study may have benefited from a prolonged monitoring period to identify the nadir of its multiple cardiovascular effects. However, the duration of angina relief with sublingual NTG is reported to be as long as 30 minutes, and such a prolongation of the protocol would not have been practical. Third, this study was conducted in patients with

mild-to-moderate epicardial disease. Consequently, extrapolation of these findings to patients with severe epicardial stenoses should be made with caution. Although prior work has shown that resting coronary flow may remain preserved even in the presence of severe stenosis,<sup>14</sup> provided microvascular function is intact, any extension of our results to this setting should be considered hypothesis-generating. In particular, the effects of NTG on epicardial resistance may differ in the presence of critical stenoses. In such lesions, where flow reserve is markedly reduced, even small degrees of vasodilatation could lead to disproportionate increases in cross-sectional area and corresponding reductions in epicardial resistance. By contrast, in patients with mild-to-moderate disease, as studied here, similar vasodilatory effects are less likely to translate into measurable changes in epicardial resistance. Fourth, the calculation of absolute flow was simplified using a validated prediction of  $T_i$ .<sup>21</sup> Whilst flow values calculated with predicted and measured  $T_i$  demonstrate excellent agreement, small differences in absolute flow values between these approaches cannot be excluded. Crucially, relative changes in flow (baseline versus post-intervention) can be considered independent of the value of  $T_i$ . Fifth, although this was a randomised, double blind, placebo-controlled study, no formal assessment of blinding was performed to verify that participants and investigators remained unaware of treatment allocation. Sixth, the original hypothesis and sample size consideration assumed an increase in resting coronary flow, whereas a decrease was observed. However, the observed reduction in absolute coronary flow of  $-8.7$  ml/min exceeded the minimal detectable difference and was estimated with reasonable precision, as reflected by a relatively narrow 95% confidence interval ( $-13.4$  to  $-3.9$  ml/min), excluding both no effect and differences close to zero. In addition, the two-sided p-value of 0.001 indicates strong statistical evidence against the null hypothesis. At this level of statistical significance, the false discovery rate, defined as the proportion of false positives among

statistically significant results at this threshold, is expected to be below 5% under conventional assumptions, supporting the robustness of the observed findings.<sup>40</sup>

## **Conclusion**

In humans, sublingual NTG induced a significant decrease in left ventricular preload and afterload, resulting in a marked decrease in myocardial work. This was associated with a significant and adaptive decrease in absolute coronary blood flow.



## **CLINICAL PERSPECTIVES**

### **What Is Known?**

Nitrates are widely used for angina and induce epicardial vasodilatation, yet their effect on absolute coronary blood flow in humans remains unclear. Their systemic effects on left ventricular preload and afterload may influence coronary haemodynamics.

### **What Is New?**

Sublingual nitroglycerin reduced left ventricular preload and afterload, lowering myocardial work. This was accompanied by increased microvascular resistance and a resulting decrease in absolute coronary blood flow, reflecting an adaptive and physiologically appropriate, demand-driven response.

### **What Is Next?**

Studies under hyperaemic or exercise conditions are needed to define the effects of nitrates when myocardial demand is increased, specifically using techniques that measure volumetric coronary flow. An integrated assessment of systemic and coronary haemodynamics may improve our understanding of this fundamental cardiovascular drug.

## **ACKNOWLEDGEMENTS**

None.

Journal Pre-proof

#### **DATA AVAILABILITY STATEMENT**

The data underlying this article will be shared on reasonable request to the corresponding author.

Journal Pre-proof

## REFERENCES

1. Münzel T, Daiber A, Gori T. Nitrate Therapy. *Circulation*. 2011;123:2132–2144.
2. Abrams J. Hemodynamic effects of nitroglycerin and long-acting nitrates. *Am Heart J*. 1985;110:216–224.
3. Smith ER, Smiseth OA, Kingma I, Manyari D, Belenkie I, Tyberg JV. Mechanism of action of nitrates. Role of changes in venous capacitance and in the left ventricular diastolic pressure-volume relation. *Am J Med*. 1984;76:14–21.
4. Garadah T, Ghaisas NK, Mehana N, Foley B, Crean P, Walsh M. Impact of intravenous nitroglycerin on pulsed Doppler indexes of left ventricular filling in acute anterior myocardial infarction. *Am Heart J*. 1998;136:812–817.
5. Bache RJ, Ball RM, Cobb FR, Rembert JC, Greenfield JC. Effects of nitroglycerin on transmural myocardial blood flow in the unanesthetized dog. *J Clin Invest*. 1975;55:1219–1228.
6. Aarnoudse W, Van't Veer M, Pijls NHJ, et al. Direct volumetric blood flow measurement in coronary arteries by thermodilution. *J Am Coll Cardiol*. 2007;50:2294–2304.
7. De Bruyne B, Pijls NHJ, Gallinoro E, et al. Microvascular Resistance Reserve for Assessment of Coronary Microvascular Function: JACC Technology Corner. *J Am Coll Cardiol*. 2021;78:1541–1549.
8. Jansen TPJ, Konst RE, Elias -Smale Suzette E., et al. Assessing Microvascular Dysfunction in Angina With Unobstructed Coronary Arteries. *J Am Coll Cardiol*. 2021;78:1471–1479.

9. Adjedj J, Picard F, Collet C, et al. Intracoronary Saline-Induced Hyperemia During Coronary Thermodilution Measurements of Absolute Coronary Blood Flow: An Animal Mechanistic Study. *J Am Heart Assoc*. 2020;9:e015793.
10. De Bruyne B, Adjedj J, Xaplanteris P, et al. Saline-Induced Coronary Hyperemia: Mechanisms and Effects on Left Ventricular Function. *Circ Cardiovasc Interv*. 2017;10:e004719.
11. Gallinoro E, Candreva A, Fernandez-Peregrina E, et al. Saline-induced coronary hyperemia with continuous intracoronary thermodilution is mediated by intravascular hemolysis. *Atherosclerosis*. 2022;352:46–52.
12. Belmonte M, Gallinoro E, Pijls NHJ, et al. Measuring Absolute Coronary Flow and Microvascular Resistance by Thermodilution: JACC Review Topic of the Week. *J Am Coll Cardiol*. 2024;83:699–709.
13. Mahendiran T, Bertolone D, Viscusi M, et al. The Influence Of Epicardial Resistance On Microvascular Resistance Reserve. *J Am Coll Cardiol*. 2024:S0735-1097(24)07198–5.
14. Mahendiran, Thabo, Collet, Carlos, De Bruyne, Bernard. Coronary-Artery Autoregulation with Increasing Stenosis. *N Engl J Med*. 2024;390:2030–2032.
15. Abbara S, Blanke P, Maroules CD, et al. SCCT guidelines for the performance and acquisition of coronary computed tomographic angiography: A report of the Society of Cardiovascular Computed Tomography Guidelines Committee: Endorsed by the North American Society for Cardiovascular Imaging (NASCI). *J Cardiovasc Comput Tomogr*. 2016;10:435–449.

16. Bermpeis K, Pauwels FA, Viscusi MM, et al. Quantification and Timing of Epicardial Vasodilation by Sublingual Nitrates. *JACC Cardiovasc Imaging*. 2024:S1936-878X(24)00230–4.
17. Torgerson DJ, Roberts C. Understanding controlled trials: Randomisation methods: concealment. *BMJ*. 1999;319:375.
18. Noonan PK, Benet LZ. Incomplete and delayed bioavailability of sublingual nitroglycerin. *Am J Cardiol*. 1985;55:184–187.
19. Johnson NP, Li W, Chen X, et al. Diastolic pressure ratio: new approach and validation vs. the instantaneous wave-free ratio. *Eur Heart J*. 2019;40:2585–2594.
20. van de Hoef TP, Lee JM, Echavarria-Pinto M, et al. Non-hyperaemic coronary pressure measurements to guide coronary interventions. *Nat Rev Cardiol*. 2020;17:629–640.
21. Mahendiran T, Fawaz S, Viscusi M, et al. Simplification of continuous intracoronary thermodilution. *EuroIntervention*. 2024;20:e1217–e1226.
22. Reddy YNV, El-Sabbagh A, Nishimura RA. Comparing Pulmonary Arterial Wedge Pressure and Left Ventricular End Diastolic Pressure for Assessment of Left-Sided Filling Pressures. *JAMA Cardiol*. 2018;3:453–454.
23. Egbe AC, Reddy YNV, Obokata M, Borlaug BA. Doppler-derived Arterial Load Indices Better Reflect Left Ventricular Afterload than Systolic Blood Pressure in Coarctation of Aorta. *Circ Cardiovasc Imaging*. 2020;13:e009672.
24. Buckberg G, Eber L, Herman M, Gorlin R. Ischemia in aortic stenosis: hemodynamic prediction. *Am J Cardiol*. 1975;35:778–784.

25. Wood SL, Crosby PA. A real-time on-line digital computer method for monitoring myocardial supply: demand ratio. *Anaesth Intensive Care*. 1981;9:28–33.
26. Asress KN, Williams R, Lockie T, et al. Physiology of Angina and Its Alleviation With Nitroglycerin. *Circulation*. 2017;136:24–34.
27. Mahendiran T, Thanou D, Senouf O, et al. AngioPy segmentation: An open-source, user-guided deep learning tool for coronary artery segmentation. *Int J Cardiol*. 2024;0.
28. Vatner S, Pagani M, Rutherford J, Millard R, Manders W. Effect of nitroglycerin on cardiac function and regional blood flow distribution in conscious dogs. *Am J Physiol*. 1978;234:H244-52.
29. Kawakami H, Sumimoto T, Hamada M, et al. Acute effect of glyceryl trinitrate on systolic blood pressure and other hemodynamic variables. *Angiology*. 1995;46:151–156.
30. Tadamura E, Mamede M, Kubo S, et al. The Effect of Nitroglycerin on Myocardial Blood Flow in Various Segments Characterized by Rest-Redistribution Thallium SPECT. *J Nucl Med*. 2003;44:745–751.
31. Gorlin R, Brachfeld N, Macleod C, Bopp P. Effect of nitroglycerin on the coronary circulation in patients with coronary artery disease or increased left ventricular work. *Circulation*. 1959;19:705–718.
32. Greenberg H, Dwyer EM, Jameson AG, Pinkernell BH. Effects of nitroglycerin on the major determinants of myocardial oxygen consumption. An angiographic and hemodynamic assessment. *Am J Cardiol*. 1975;36:426–432.

33. Bernstein L, Friesinger GC, Lichtlen PR, Ross RS. The effect of nitroglycerin on the systemic and coronary circulation in man and dogs: myocardial blood flow measured with xenon. *Circulation*. 1966;33:107–116.
34. Mahendiran T, Corradetti S, Perera D, Pijls NHJ, De Bruyne B. Coronary autoregulation: the complex interplay between epicardial and microvascular resistance. *Eur Heart J*. 2026;ehag145.
35. Bond GS. Effect of various agents on the blood flow through the coronary arteries and veins. *J Exp Med*. 1910;12:575–585.
36. Mahendiran T, Wilgenhof A, De Bruyne B. The Elusiveness of “Rest”: The Challenges of Assessing Resting Coronary Physiology. *Circulation*. 2025;152:147–149.
37. Parker JO, Augustine RJ, Burton JR, West RO, Armstrong PW. Effect of nitroglycerin ointment on the clinical and hemodynamic response to exercise. *Am J Cardiol*. 1976;38:162–166.
38. Mahmarian JJ, Fenimore NL, Marks GF, et al. Transdermal nitroglycerin patch therapy reduces the extent of exercise-induced myocardial ischemia: results of a double-blind, placebo-controlled trial using quantitative thallium-201 tomography. *J Am Coll Cardiol*. 1994;24:25–32.
39. Fallen EL, Nahmias C, Scheffel A, Coates G, Beanlands R, Garnett ES. Redistribution of myocardial blood flow with topical nitroglycerin in patients with coronary artery disease. *Circulation*. 1995;91:1381–1388.



40. Jüni P, Antoniou S, Arbelo E, et al. 2024 Revision of the level of evidence grading system for ESC clinical practice guideline recommendations I: therapy and prevention. *Eur Heart J*. 2025;46:1885–1894.

**Table 1. Baseline characteristics.** BMI = body mass index; MI = myocardial infarction; PCI = percutaneous coronary intervention; ACEi = Angiotensin-converting enzyme inhibitor; ARB = angiotensin receptor blockers; eGFR = estimated glomerular filtration rate; LVEF = left ventricular ejection fraction.

| Characteristic                         | NTG (n=20)   | Placebo (n=20) |
|----------------------------------------|--------------|----------------|
| <b><i>Clinical and demographic</i></b> |              |                |
| Age, mean (SD)                         | 65.8 (11.8)  | 65.7 (9.0)     |
| Male, n (%)                            | 17 (85.0)    | 14 (70.0)      |
| Hypertension, n (%)                    | 16 (80.0)    | 13 (65.0)      |
| Dyslipidaemia, n (%)                   | 16 (80.0)    | 18 (90.0)      |
| Diabetes, n (%)                        | 4 (20.0)     | 4 (20.0)       |
| BMI, mean (SD)                         | 28.2 (5.2)   | 27.8 (3.8)     |
| Active smoking, n (%)                  | 1 (5.0)      | 2 (10.0)       |
| Family history of CAD, n (%)           | 5 (25.0)     | 4 (20.0)       |
| Previous MI, n (%)                     | 2 (10.0)     | 4 (20.0)       |
| Previous PCI, n (%)                    | 3 (15.0)     | 5 (25.0)       |
| LVEF, mean (SD)                        | 55.5 (6.7)   | 55.6 (4.8)     |
| eGFR, mean (SD)                        | 77.3 (16.1)  | 82.8 (9.5)     |
| <b><i>Medication</i></b>               |              |                |
| Aspirin, n (%)                         | 10 (50.0)    | 9 (45.0)       |
| ACEi/ARB, n (%)                        | 9 (45.0)     | 11 (55.0)      |
| Betablocker, n (%)                     | 10 (50.0)    | 11 (55.0)      |
| Calcium channel blocker, n (%)         | 4 (20.0)     | 3 (15.0)       |
| Oral anticoagulation, n (%)            | 4 (20.0)     | 4 (20.0)       |
| Statin, n (%)                          | 15 (75.0)    | 14 (70.0)      |
| Oral anti-diabetic, n (%)              | 4 (20.0)     | 3 (15.0)       |
| <b><i>Clinical presentation</i></b>    |              |                |
| Chest pain, n (%)                      | 13 (65.0)    | 13 (65.0)      |
| Dyspnoea, n (%)                        | 6 (30.0)     | 7 (35.0)       |
| Abnormal functional testing, n (%)     | 2 (10.0)     | 2 (10.0)       |
| <b><i>Haemodynamic parameters</i></b>  |              |                |
| LVEDP (mmHg)*                          | 14.4 (5.8)   | 16.4 (7.7)     |
| P <sub>a</sub> (mmHg)                  | 101.2 (10.9) | 105.1 (15.2)   |

|                                     |                |                |
|-------------------------------------|----------------|----------------|
| $P_d$ (mmHg)                        | 91.1 (11.5)    | 94.0 (17.8)    |
| $P_d / P_a$                         | 0.89 (0.09)    | 0.90 (0.03)    |
| SPTI (mmHg•s)                       | 41.7 (5.6)     | 43.3 (7.9)     |
| LV work/minute<br>(mmHg•s/min)      | 2886.6 (537.3) | 3116.6 (804.5) |
| Buckberg index                      | 1.2 (0.3)      | 1.1 (0.3)      |
| Vessel volume<br>(mm <sup>3</sup> ) | 481.5 (161.4)  | 475.9 (193.7)  |
| $R_{epi}$ (WU)                      | 179.5 (74.2)   | 197.8 (179.8)  |
| $R_{\mu}$ (WU)                      | 1592.8 (408.3) | 1559.7 (432.7) |
| Q (ml/min)                          | 60.9 (16.9)    | 64.5 (22.2)    |

**Table 2. Summary of measured haemodynamic parameters.** For each outcome, the estimated mean difference between groups was analysed using analysis of covariance (ANCOVA) adjusted for the outcomes's baseline values. LVEDP = left ventricular end-diastolic pressure;  $P_a$  = aortic pressure;  $P_d$  = distal coronary pressure; Q = resting absolute coronary flow;  $R_{epi}$  = absolute resting epicardial resistance;  $R_{\mu}$  = absolute resting microvascular resistance; SPTI = systolic pressure-time index. \*The repeat measurement of LVEDP was performed at the end of the protocol (at 12 minutes) with a pigtail catheter. \*\*Vessel volumes were recalculated using 3D-QCA with repeat images acquired at the end of the protocol.

| Parameter                               | NTG              | Placebo          | Estimated mean difference (95% CI) | p value | Percentage change (95% CI) |
|-----------------------------------------|------------------|------------------|------------------------------------|---------|----------------------------|
| <b>LVEDP (mmHg)*</b>                    |                  |                  |                                    |         |                            |
| 12 minutes                              | 10.1±6.2         | 14.2±5.1         | -5.4 (-7.8 to -3.1)                | <0.001  | -38.1 (-54.4 to -21.8)     |
| <b><math>P_a</math> (mmHg)</b>          |                  |                  |                                    |         |                            |
| 5 minutes                               | 99.4±14.3        | 100.6±11.3       | -4.7 (-7.5 to -2.0)                | 0.001   | -4.7 (-7.1 to -2.2)        |
| 10 minutes                              | 95.9±14.1        | 101.2±11.5       | -8.6 (-12.2 to -5.1)               | <0.001  | -8.5 (-11.7 to -5.3)       |
| <b><math>P_d</math> (mmHg)</b>          |                  |                  |                                    |         |                            |
| 5 minutes                               | 89.7±16.0        | 90.7±11.8        | -3.6 (-6.0 to -1.1)                | 0.006   | -3.7 (-6.3 to -1)          |
| 10 minutes                              | 86.7±15.3        | 91.3±12.4        | -7.1 (-10.6 to -3.7)               | <0.001  | -7.4 (-11.1 to -3.7)       |
| <b><math>P_d/P_a</math></b>             |                  |                  |                                    |         |                            |
| 5 minutes                               | <b>0.90±0.08</b> | <b>0.90±0.03</b> | 0.01 (0.00 to 0.01)                | 0.004   | 1.0 (0.2 to 1.8)           |
| 10 minutes                              | <b>0.90±0.07</b> | <b>0.90±0.03</b> | 0.01 (0.00 to 0.02)                | 0.026   | 1.2 (-0.1 to 2.6)          |
| <b>SPTI (mmHg*s)</b>                    |                  |                  |                                    |         |                            |
| 5 minutes                               | 36.8±7.7         | 41.2±5.9         | -5.8 (-8.0 to -3.7)                | <0.001  | -14.1 (-18.8 to -9.4)      |
| 10 minutes                              | 34.3±7.2         | 41.9±6.1         | -8.9 (-11.6 to -6.2)               | <0.001  | -21.1 (-26.9 to -15.4)     |
| <b>LV work/minute (mmHg*s/min)</b>      |                  |                  |                                    |         |                            |
| 5 minutes                               | 2588.7±580.1     | 2758.0±492.9     | -319.8 (-516.7 to -122.9)          | 0.002   | -11.6 (-17.5 to -5.8)      |
| 10 minutes                              | 2408.1±504.1     | 2797.1±482.1     | -515.8 (-725.3 to -306.3)          | <0.001  | -18.5 (-25.2 to -11.8)     |
| <b>Buckberg index</b>                   |                  |                  |                                    |         |                            |
| 5 minutes                               | 1.4±0.4          | 1.2±0.3          | 0.2 (0.1 to 0.3)                   | 0.001   | 18.6 (7.5 to 29.7)         |
| 10 minutes                              | 1.4±0.4          | 1.3±0.3          | 0.2 (0.1 to 0.3)                   | <0.001  | 17.7 (9.3 to 26.1)         |
| <b>Vessel volume (mm<sup>3</sup>)**</b> |                  |                  |                                    |         |                            |
| 10 minutes                              | 559.6±198.3      | 483.3±195.6      | 64.1 (1.1 to 127.2)                | 0.046   | 15.6 (0.9 to 30.2)         |

|                             |              |              |                       |       |                       |
|-----------------------------|--------------|--------------|-----------------------|-------|-----------------------|
| <b>R<sub>epi</sub> (WU)</b> |              |              |                       |       |                       |
| 5 minutes                   | 190.2±177.9  | 178.0±71.6   | -5.7 (-17.0 to 5.7)   | 0.317 | -5 (-12.7 to 2.7)     |
| 10 minutes                  | 190.5±162.0  | 177.3±75.4   | -3.2 (-18.6 to 12.1)  | 0.673 | -0.5 (-11.2 to 10.2)  |
| <b>R<sub>μ</sub> (WU)</b>   |              |              |                       |       |                       |
| 5 minutes                   | 1645.0±450.1 | 1610.0±403.4 | 65.7 (-47.5 to 178.8) | 0.247 | 4.5 (-3 to 12.1)      |
| 10 minutes                  | 1686.8±490.6 | 1598.0±398.1 | 121.5 (15.3 to 227.8) | 0.026 | 8.0 (1.1 to 14.9)     |
| <b>Q (ml/min)</b>           |              |              |                       |       |                       |
| 5 minutes                   | 57.7±17.1    | 59.7±16.3    | -4.7 (-9.6 to 0.1)    | 0.054 | -7.6 (-15.1 to -0.2)  |
| 10 minutes                  | 54.5±15.6    | 60.6±16.4    | -8.7 (-13.4 to -3.9)  | 0.001 | -13.3 (-20.2 to -6.4) |

## FIGURE LEGENDS

**Figure 1. Study protocol summary.** The protocol consisted of three separate continuous intracoronary thermodilution measurement windows. Baseline measurements were taken for 1 minute before the intervention was given. Measurements were repeated at 5 minutes and 10 minutes. DPTI = diastolic pressure-time index; LVEDP = left ventricular end-diastolic pressure;  $P_a$  = aortic pressure;  $P_d$  = distal coronary pressure;  $Q_{rest}$  = resting absolute coronary flow;  $R_{\mu,rest}$  = absolute resting microvascular resistance; SPTI = systolic pressure-time index; Buckberg = Buckberg index (i.e. DPTI/SPTI).

**Figure 2. Example pressure and temperature tracings.** (A) a patient from the placebo group, and (B) a patient from the NTG group. Data were recorded and exported from CoroFlow Cardiovascular System (Coroventis).  $P_a$  = aortic pressure (red);  $P_d$  = distal coronary pressure (green),  $Q_{rest}$  = resting absolute coronary flow; T = mixed blood and saline temperature measured in the distal coronary artery (blue).

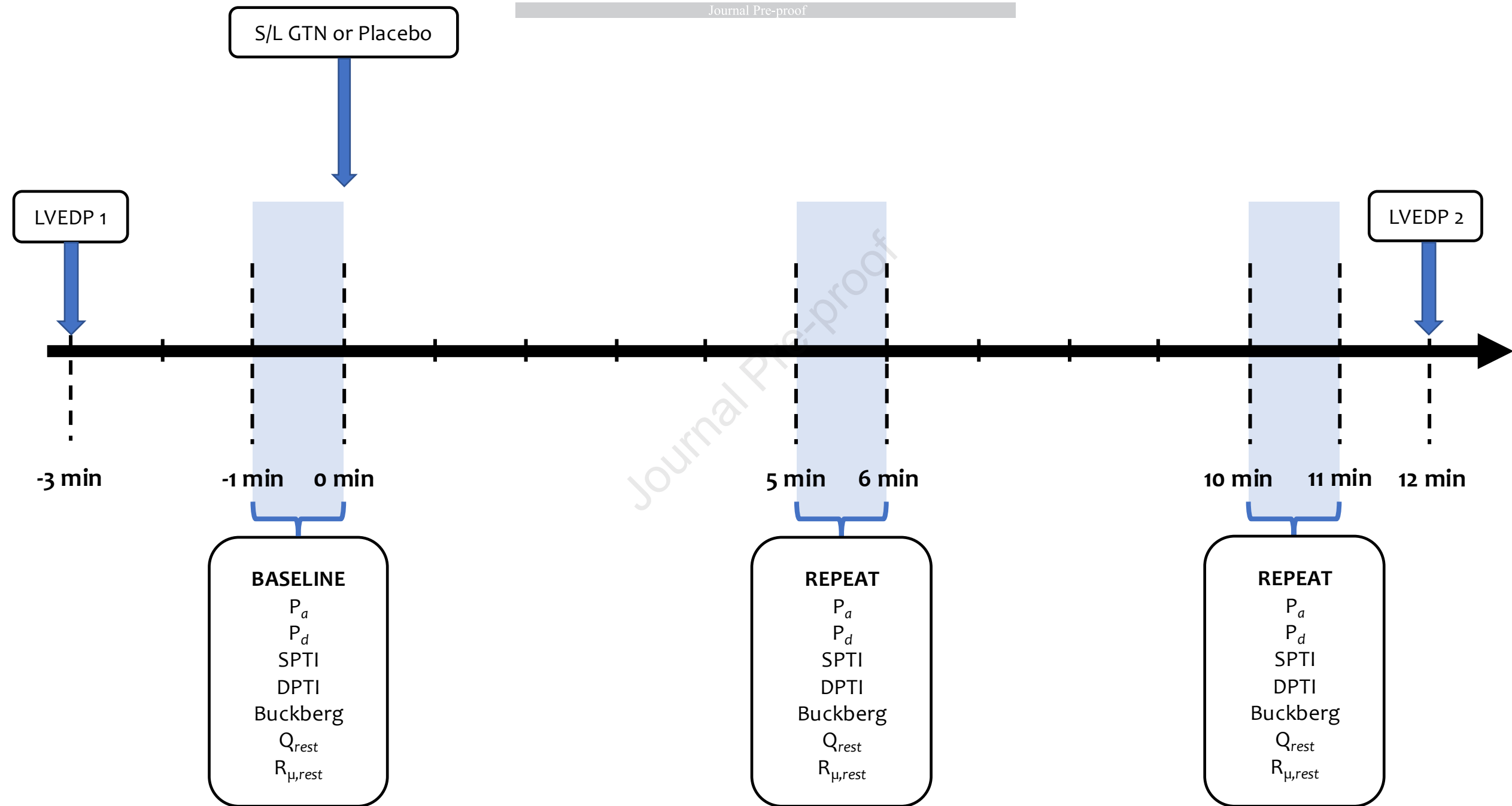
**Figure 3. Effect of sublingual NTG on left ventricular preload.** Changes in left ventricular end-diastolic pressure (LVEDP) after the administration of sublingual NTG. Shown are the mean percentage changes from baseline, with 95% confidence intervals. P values reflect between-group differences assessed by analysis of covariance (ANCOVA) as shown in **Table 2**.

**Figure 4. Effect of sublingual NTG on left ventricular afterload.** The impact of NTG on (A) aortic pressure ( $P_a$ ) and the consequential effect on (B) distal coronary pressure ( $P_d$ ). Shown are the mean percentage change from baseline with 95% confidence intervals. P values reflect between-group differences assessed by analysis of covariance (ANCOVA) as shown in **Table 2**.

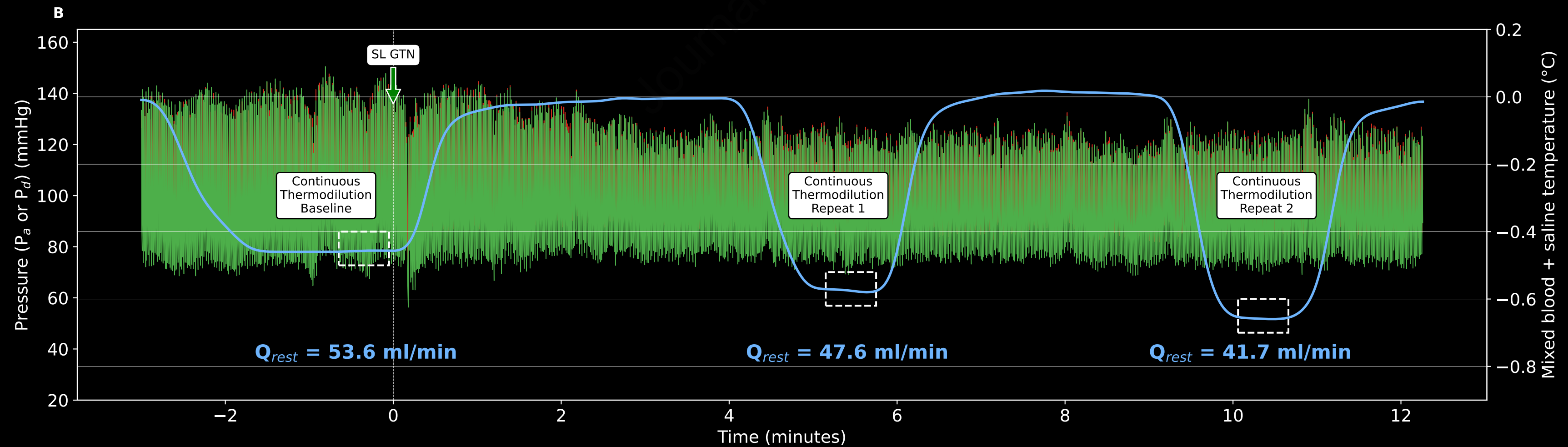
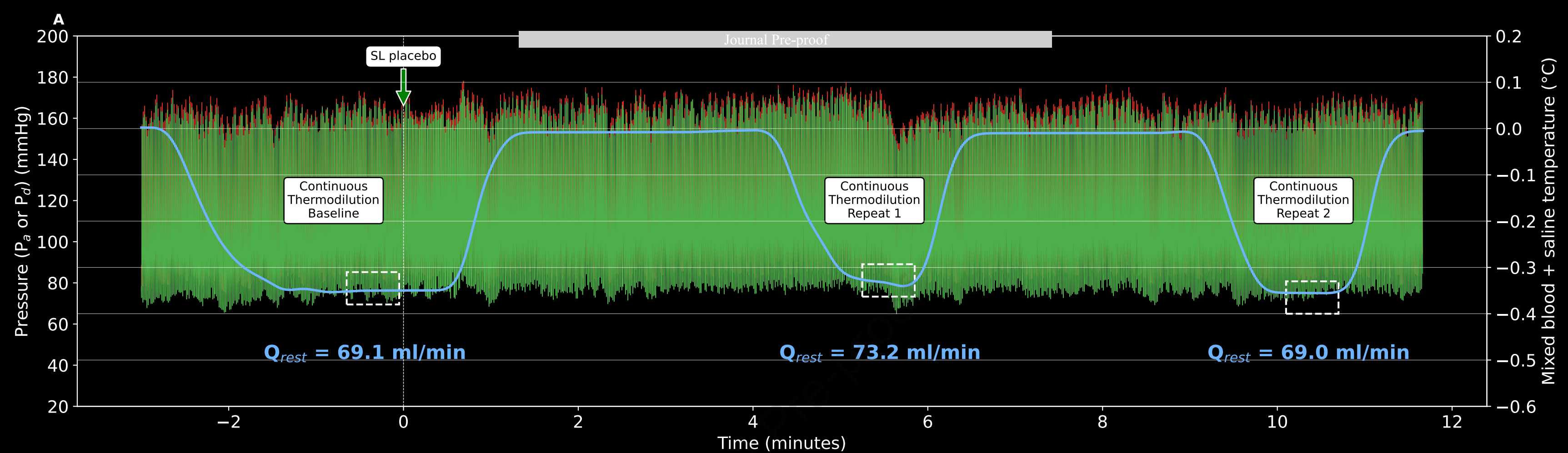
**Figure 5. Effect of sublingual NTG on myocardial work and perfusion.** The impact of NTG on the (A) systolic pressure-time index (SPTI); (B) left ventricular work per minute (LVW/min); and (C) Buckberg index. Shown are the mean percentage changes from baseline with 95% confidence intervals. P values reflect between-group differences assessed by analysis of covariance (ANCOVA) as shown in **Table 2**.

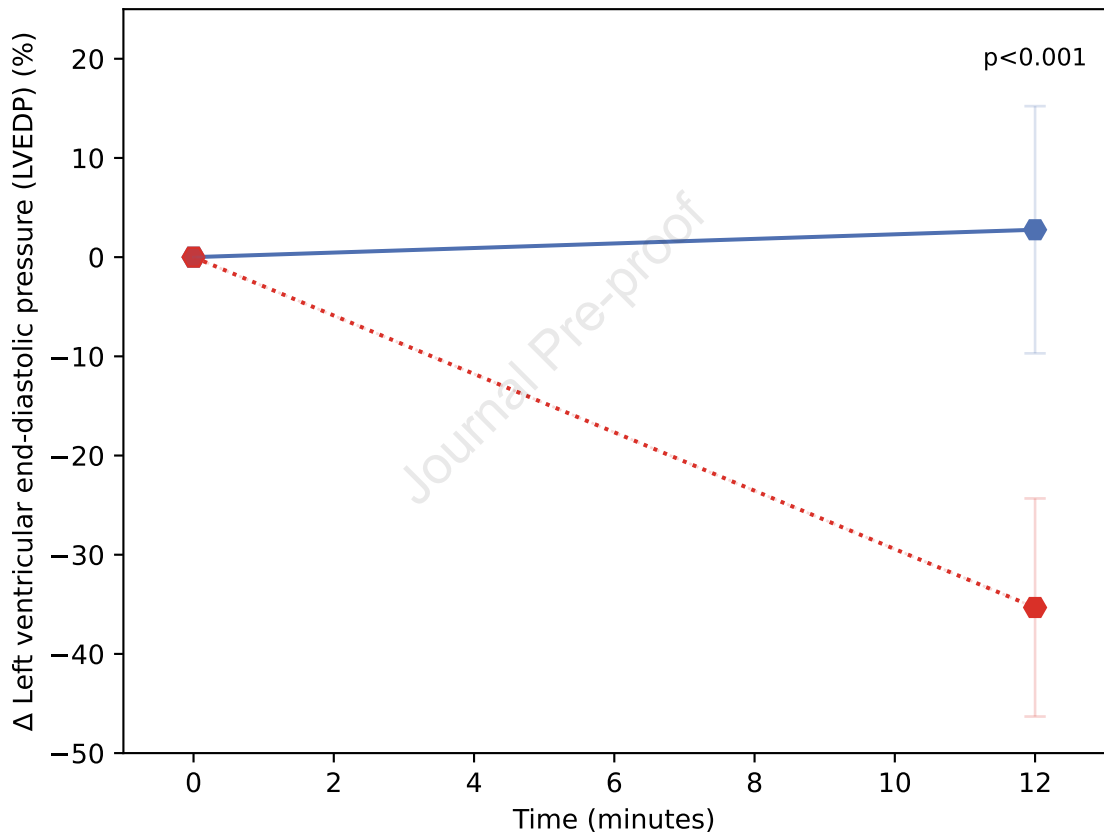
**Figure 6. Effect of sublingual NTG on microvascular resistance and absolute coronary flow.** The impact of NTG on (A) absolute resting microvascular resistance ( $R_{\mu,rest}$ ), and (B) absolute resting coronary blood flow ( $Q_{rest}$ ). Shown are the mean percentage changes from baseline with 95% confidence intervals. P values reflect between-group differences assessed by analysis of covariance (ANCOVA) as shown in **Table 2**.

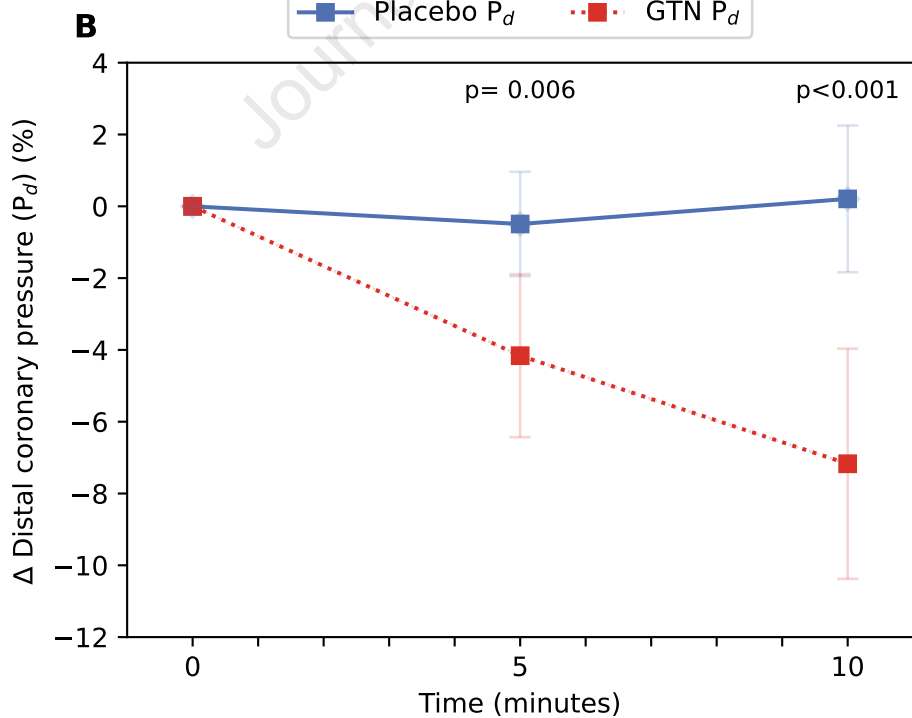
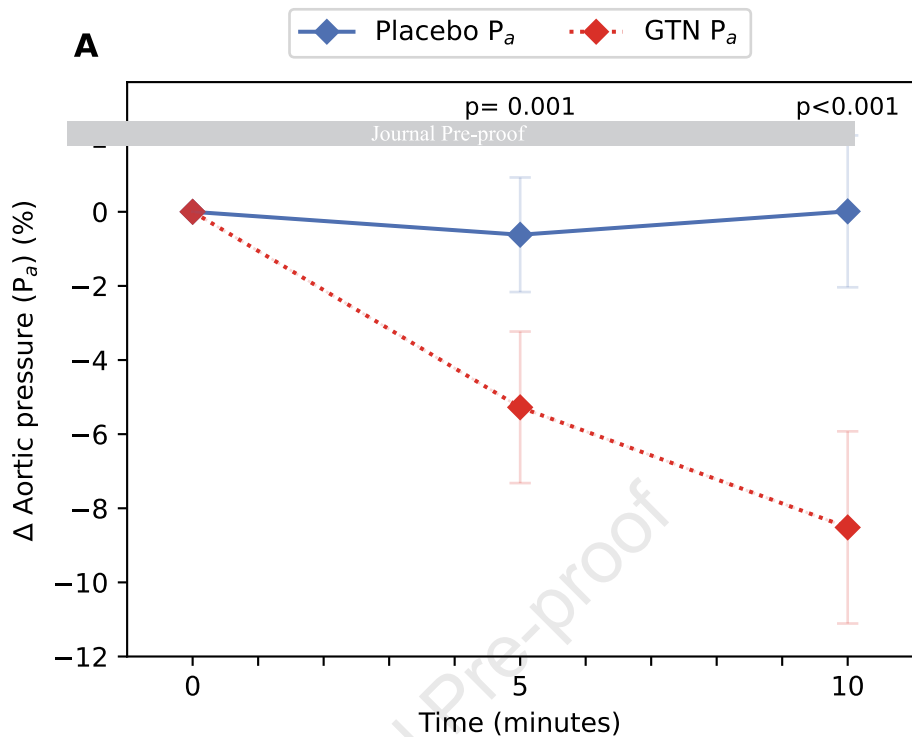
**Central Illustration. The effect of sublingual nitroglycerin on absolute coronary blood flow.** In a randomised, double-blind, placebo-controlled mechanistic study, 40 patients with suspected angina underwent continuous intracoronary thermodilution before, and then 5 and 10 minutes after sublingual nitroglycerin (NTG) or placebo. NTG reduced LVEDP, aortic and distal coronary pressures, and myocardial work. Despite increased epicardial vessel volume, NTG increased microvascular resistance, resulting in a decrease in absolute coronary blood flow of 13.3%. These findings suggest that sublingual NTG lowers preload and afterload, reducing myocardial work and producing an adaptive reduction in absolute coronary blood flow. NTG = nitroglycerin; LV = left ventricle; LVEDP = left ventricular end-diastolic pressure

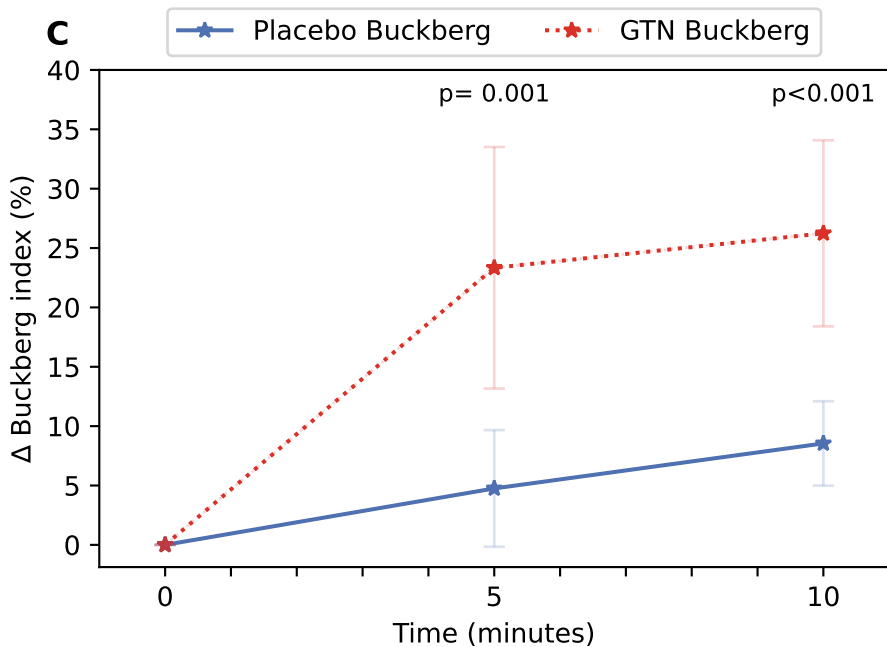
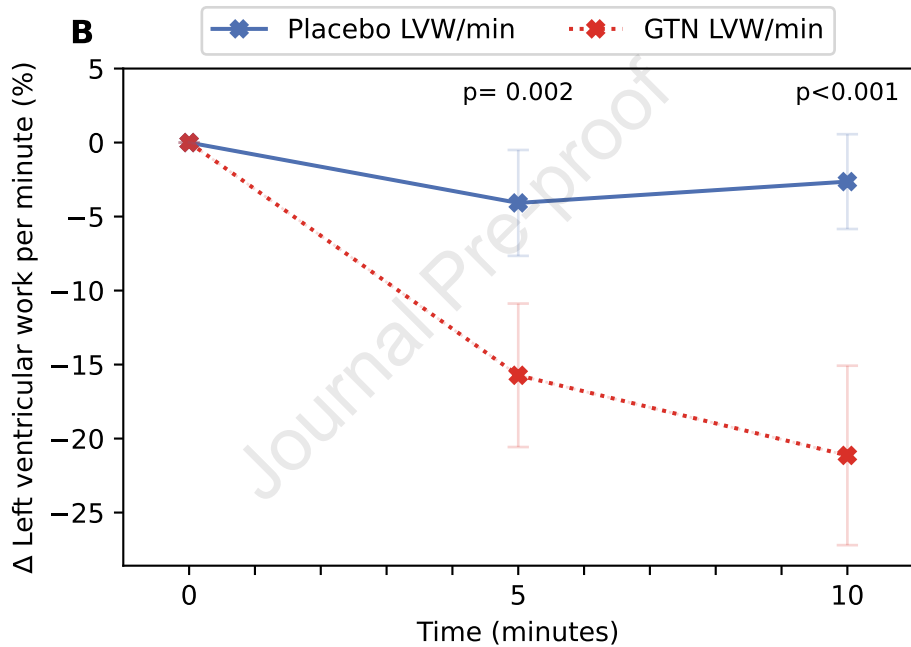
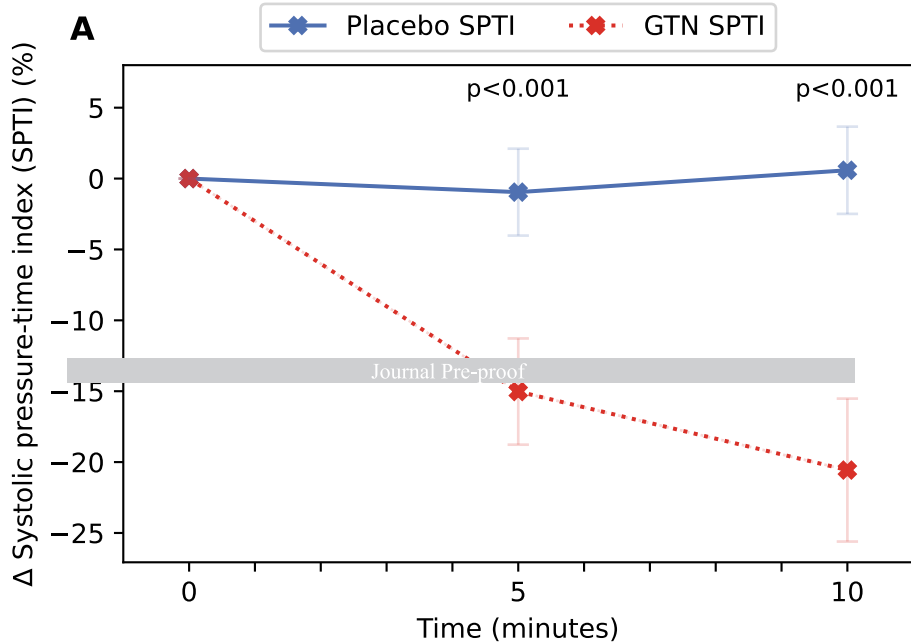


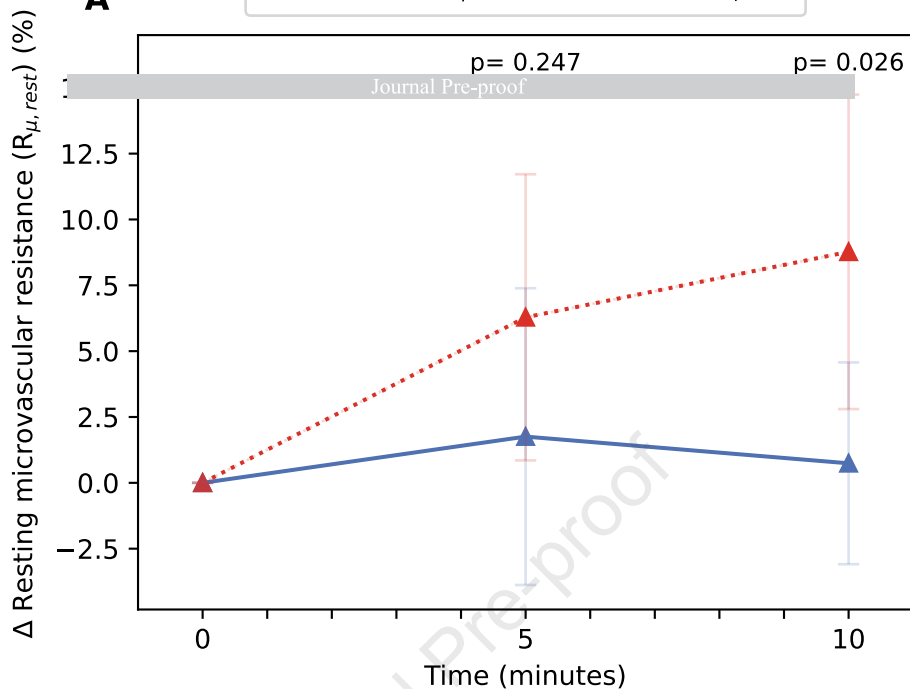










**A**Placebo  $R_{\mu, rest}$  GTN  $R_{\mu, rest}$ **B**

Placebo Q GTN Q

